

Statement of Work (SOW)

Title: Boulder Micro-Fabrication Facility (BMF) Tystar Furnace Service Contract

I. Background Information

The National Institute of Standards and Technology (NIST) Boulder location has a state-of-the-art research laboratory (the Katharine Blodgett Gebbie Building) to support the most sensitive research performed at NIST. Included in this building is a Class 100 microelectronics fabrication facility, called the Boulder Micro-Fabrication Facility (BMF). Among the many critical pieces of equipment in this laboratory is the Tystar MiniTytan 4600 furnace. This tool is a critical piece of the BMF infrastructure used to deposit high-quality dielectric films onto substrates. To keep a high tool uptime, the BMF requires a service contract to maintain this tool and offer support if the tool becomes inoperable.

Purpose/Objective: The purpose of this requirement is to provide a service contract for the BMF's Tystar MiniTytan 4600 furnace for atmospheric and Low-Pressure Chemical Vapor Deposition (LPCVD) tube changes, annual Preventative Maintenance (PM), and emergency maintenance visits.

II. Scope

The Contractor shall provide a service maintenance agreement for one (1) each Tystar MiniTytan 4600 furnace to include all planned materials, consumables, travel, labor, equipment, and expenses.

III. Specific Requirements:

Line Item 0001:

Description: Preventative Maintenance Service (Base Period)

Quantity: 1 LOT

The contractor shall:

1. Provide a preventive maintenance (PM) service on the existing Tystar MiniTytan 4600 furnace for each of the four tubes and the furnace as a whole except rebuilding or replacing the vacuum pumps and include all of the steps as required per the "Tytan Preventive Maintenance Procedure". All services shall be provided at a mutually agreeable time between the Government and the Contractor.
2. Provide a backup of the configuration and recipe files.
3. Perform the PM service in a method that is compatible with an ISO class 5 cleanroom.
4. Provide all parts and labor required for the PM service.
5. After completion of the PM service, ensure that the tool be able to operate according to OEM specifications or better.
6. Within one week of each service visit, provide a report detailing all of the services that were performed.

The following line items are option line items/optional tasks and may be exercised in accordance with the Option Clause included herein.

Line Item 0002: Optional Task

Description: Atmospheric Tube Replacement (Base Period)

Quantity: 1 LOT

At a time mutually coordinated by the Government and the Contractor, the Contractor shall provide an atmospheric tube replacement service for the Government's existing Tystar MiniTytan 4600 furnace, and shall include the following:

1. Replace the process tube according to OEM guidelines including replacing the following parts with OEM parts:
 - a. process tube
 - b. cantilever sheathes
 - c. thermocouple sheath
 - d. all related gaskets and O-rings as defined by the OEM
2. Be performed in a method that is compatible with an ISO class 5 cleanroom.
3. Include all parts and labor required for this atmospheric tube replacement service.
4. After completion, be able to operate according to OEM specifications or better.

Within one week of each service visit, provide a report detailing all of the services that were performed.

Line Item 0003: Optional Task

Description: LPCVD Tube Replacement (Base Period)

Quantity: 1 LOT

At a time mutually coordinated by the Government and the Contractor, the Contractor shall provide a LPCVD tube replacement service on the Government's Tystar MiniTytan 4600 furnace, and shall include the following:

1. Replace the process tube according to OEM guidelines including replacing the following parts with OEM parts:
 - a. process tube
 - b. cantilever sheathes
 - c. thermocouple sheath
 - d. all related gaskets and O-rings as defined by the OEM
2. Be performed in a method that is compatible with an ISO class 5 cleanroom.
3. Provide all parts and labor required for this atmospheric tube replacement service.
4. After completion of the atmospheric tube replacement service, the tool must be able to operate according to OEM specifications or better.
5. Within one week of each service visit, the contractor shall provide a report detailing all of the services that were performed.

Line Item 0004: Optional Task

Description: Emergency Maintenance and Repair (Base Period)
Quantity: 1 LOT

The Contractor shall provide all necessary parts, labor, travel, and related resources required to maintain or repair the system, ensuring it meets or exceeds original equipment manufacturer (OEM) specifications.

Line Item 1001: OPTION LINE ITEM
Description: Preventative Maintenance Service (Option Period 1)
Quantity: 1 LOT

The contractor shall provide a PM service as described in CLIN 0001.

Line Item 1002: Optional Task
Description: Atmospheric Tube Replacement (Option Period 1)
Quantity: 1 LOT

The contractor shall provide an atmospheric tube replacement service as described in CLIN 0002.

Line Item 1003: Optional Task
Description: LPCVD Tube Replacement (Option Period 1)
Quantity: 1 LOT

The contractor shall provide an atmospheric tube replacement service as described in CLIN 0003.

Line Item 1004: Optional Task
Description: Emergency Maintenance and Repair (Option Period 1)
Quantity: 1 LOT

The Contractor shall provide all necessary parts, labor, travel, and related resources required to maintain or repair the system, ensuring it meets or exceeds original equipment manufacturer (OEM) specifications.

Line Item 2001: OPTION LINE ITEM
Description: Preventative Maintenance Service (Option Period 2)
Quantity: 1 LOT

The contractor shall provide a PM service as described in CLIN 0001.

Line Item 2002: Optional Task
Description: Atmospheric Tube Replacement (Option Period 2)
Quantity: 1 LOT

The contractor shall provide an atmospheric tube replacement service as described in CLIN 0002

Line Item 2003: Optional Task

Description: LPCVD Tube Replacement (Option Period 2)

Quantity: 1 LOT

The contractor shall provide an atmospheric tube replacement service as described in CLIN 0003.

Line Item 2004: Optional Task

Description: Emergency Maintenance and Repair (Option Period 2)

Quantity: 1 LOT

The Contractor shall provide all necessary parts, labor, travel, and related resources required to maintain or repair the system, ensuring it meets or exceeds original equipment manufacturer (OEM) specifications.

Line Item 3001: OPTION LINE ITEM

Description: Preventative Maintenance Service (Option Period 3)

Quantity: 1 LOT

The contractor shall provide a PM service as described in CLIN 0001.

Line Item 3002: Optional Task

Description: Atmospheric Tube Replacement (Option Period 3)

Quantity: 1 LOT

The contractor shall provide an atmospheric tube replacement service as described in CLIN 0002.

Line Item 3003: Optional Task

Description: LPCVD Tube Replacement (Option Period 3)

Quantity: 1 LOT

The contractor shall provide an atmospheric tube replacement service as described in CLIN 0003.

Line Item 3004: Optional Task

Description: Emergency Maintenance and Repair (Option Period 3)

Quantity: 1 LOT

The Contractor shall provide all necessary parts, labor, travel, and related resources required to maintain or repair the system, ensuring it meets or exceeds original equipment manufacturer (OEM) specifications.

Line Item 4001: OPTION LINE ITEM

Description: Preventative Maintenance Service (Option Period 4)

Quantity: 1 LOT

The contractor shall provide a PM service described in CLIN 0001.

Line Item 4002: Optional Task

Description: Atmospheric Tube Replacement (Option Period 4)

Quantity: 1 LOT

The contractor shall provide an atmospheric tube replacement service described in CLIN 0002.

Line Item 4003: Optional Task

Description: LPCVD Tube Replacement (Option Period 4)

Quantity: 1 LOT

The contractor shall provide an atmospheric tube replacement service described in CLIN 0003.

Line Item 4004: Optional Task

Description: Emergency Maintenance and Repair (Option Period 4)

Quantity: 1 LOT

The Contractor shall provide all necessary parts, labor, travel, and related resources required to maintain or repair the system, ensuring it meets or exceeds original equipment manufacturer (OEM) specifications.

IV. Deliverables

<i>Description</i>	<i>Quantity or Format</i>	<i>Due Date</i>
Report detailing all of the services that are performed.	1 EACH Include backup of the configuration and recipe files.	Within one week of each service visit

Standards of Acceptance: The NIST TPOC or COR shall review (list deliverable) and respond with an acceptance or request for revision email to the Contractor Point of Contact (POC) within 5 business days of receipt of deliverable. This section applies to deliverables which must be reviewed and approved for acceptances (draft plans, drawings, etc.)

V. Place of Performance:

The place of performance shall be at NIST 325 Broadway, Boulder, Colorado location.

Access to Building 81 shall be granted to the contractor for the entire period of performance of this contract.

VI. Period of Performance:

The period of performance shall be a base period of twelve (12) months plus four (4) 12-month option periods. The estimated start date is June 1, 2026.

VII. Government Furnished Property:

a. N/A

VIII. Performance Requirement Summary

Task	Performance Indicators	Performance Standards	Acceptable Quality Level	Surveillance Method
The contractor shall provide a service contract to the BMF's Tystar MiniTytan 4600 furnace according to the specifications in Section 2.	The BMF's Tystar MiniTytan 4600 operates at OEM specifications or better and if the tool is inoperable, the contractor provides emergency maintenance in accordance with the guidelines in section 2.c.	The technical requirements defined in the statement of work section 2 are performed in a timely manner as defined in section 2.c.	The BMF's Tystar MiniTytan 4600 furnace operates at OEM specifications or better. Repairs or replacements are completed with OEM parts by factory trained staff.	Visual verification of the completed work and the Tystar MiniTytan 4600 furnace operates at OEM specifications or better.

IX. Travel:

The contractor shall be responsible for all travel expenses.

X. Contractor Supplied Materials

The contractor shall supply all materials required for the service and maintenance of the Tystar MiniTytan 4600 furnace.

XI. Payment Schedule

Advance payment is not authorized. The Contractor must invoice in arrears according to the payment schedule. The Contractor shall be paid, in accordance with Net 30-day payment terms, upon receipt and acceptance of a proper invoice.

XII. Experience Requirements

Contractor's Minimum Qualifications

The contractor shall:

- i. Provide personnel that have been factory trained in the service and maintenance of a Tystar MiniTytan 4600 furnace by the original manufacturer (Tystar).

- ii. Provide personnel that pass required government background checks to be issued badges for work on site.
- iii. Provide personnel that have been factory trained in the service and maintenance of a Tystar MiniTytan 4600 by the original manufacturer (Tystar). This condition shall be supported by appropriate documentation.
- iv. Provide personnel that pass required government background checks to be issued badges for work on site.

NOTE: All staff must be badged by NIST Security. No one may enter the site without a badge. The work shall be scheduled with the TPOC / COR prior to arrival to do the work.

The day-to-day supervision and direct control over the work performed by the on-site contractor personnel shall be the sole responsibility of the contractor.

XIII. Miscellaneous Information

Safety: The Contractor employee must be responsible for knowing and complying with all installation safety prevention regulations. Such regulations include, but are not limited to, general safety, fire prevention, and waste disposal.

Security: NIST is a restricted campus. An identification badge is required for access for entry into buildings and is shown to the armed Security Police when entering the campus.

Regular Business Hours. Regular business hours are Monday through Friday, 9:00 am to 5:00 pm Mountain Time, excluding Federal holidays and NIST closures.

Identification Badges: Contractor employees must comply with NIST identification and access requirements. Each Contractor employee must wear a visible identification badge provided by the NIST Security Office.

Vehicle Registration: All Contractor employees must register their vehicles with the NIST Security Office to gain access to the campus. A valid driver's license, Government-furnished civilian ID, proof of insurance and current registration must be presented to the NIST Security Office, at which time a NIST vehicle pass will be issued. The pass must be displayed on the vehicle in accordance with NIST Security Office instructions.